

TUGAS JARKOM PRAKTIKUM WEEK 12

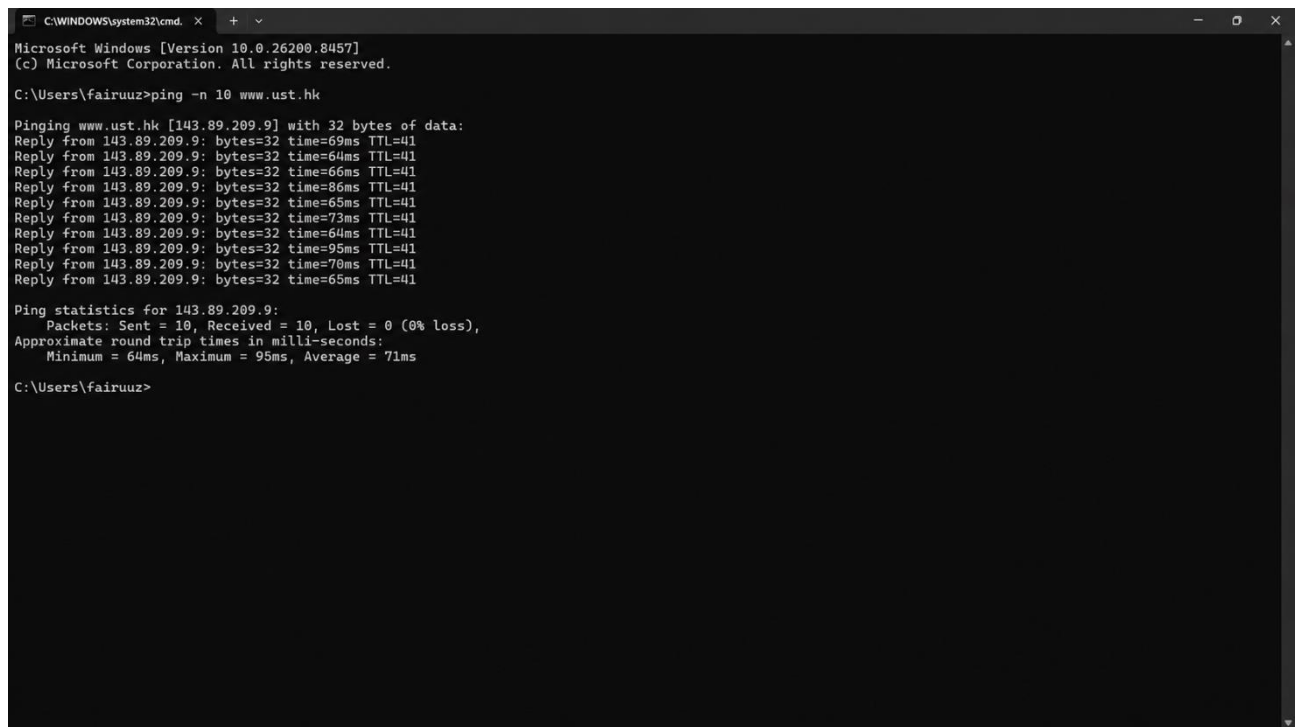
ICMP

Tujuan :

1. Mahasiswa dapat menginvestigasi cara kerja protokol ICMP menggunakan Wireshark
2. Mahasiswa dapat membuat program ICMP Pinger

ICMP dan PING

1. ping -n 10 www.ust.hk



```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\fairuuz>ping -n 10 www.ust.hk

Pinging www.ust.hk [143.89.209.9] with 32 bytes of data:
Reply from 143.89.209.9: bytes=32 time=69ms TTL=41
Reply from 143.89.209.9: bytes=32 time=64ms TTL=41
Reply from 143.89.209.9: bytes=32 time=66ms TTL=41
Reply from 143.89.209.9: bytes=32 time=86ms TTL=41
Reply from 143.89.209.9: bytes=32 time=65ms TTL=41
Reply from 143.89.209.9: bytes=32 time=73ms TTL=41
Reply from 143.89.209.9: bytes=32 time=64ms TTL=41
Reply from 143.89.209.9: bytes=32 time=95ms TTL=41
Reply from 143.89.209.9: bytes=32 time=70ms TTL=41
Reply from 143.89.209.9: bytes=32 time=65ms TTL=41

Ping statistics for 143.89.209.9:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 64ms, Maximum = 95ms, Average = 71ms

C:\Users\fairuuz>
```

2. ICMP (Echo Ping Request)

The screenshot shows a Wireshark packet capture of an ICMP Echo (ping) request. The packet list pane displays a table of captured packets, with packet 260 selected. The packet details pane shows the structure of the selected packet, and the packet bytes pane displays the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
260	11.6241095	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 261)

Frame 260: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface
Ethernet II, Src: LiteonTechno_af:cf:7d (24:b2:b9:af:cf:7d), Dst: TPLink_5b:1b:b4 (bc:07:1d:5b:1b:b4)
Internet Protocol Version 4, Src: 192.168.0.102, Dst: 143.89.209.9
Internet Control Message Protocol
Type: Echo (ping) request (8)
Code: 0
Checksum: 0x4d59 [correct]
[Checksum Status: Good]
Identifier (BE): 1 (0x0001)
Identifier (LE): 256 (0x0100)
Sequence Number (BE): 2 (0x0002)
Sequence Number (LE): 512 (0x0200)
[Request frame: 261]
Data (32 bytes)

3. ICMP (Echo Ping Reply)

The screenshot shows a Wireshark packet capture of an ICMP Echo (ping) reply. The packet list pane displays a table of captured packets, with packet 261 selected. The packet details pane shows the structure of the selected packet, and the packet bytes pane displays the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
261	11.6938367	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=41 (request in 260)

Frame 261: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface
Ethernet II, Src: TPLink_5b:1b:b4 (bc:07:1d:5b:1b:b4), Dst: LiteonTechno_af:cf:7d (24:b2:b9:af:cf:7d)
Internet Protocol Version 4, Src: 143.89.209.9, Dst: 192.168.0.102
Internet Control Message Protocol
Type: Echo (ping) reply (0)
Code: 0
Checksum: 0x5559 [correct]
[Checksum Status: Good]
Identifier (BE): 1 (0x0001)
Identifier (LE): 256 (0x0100)
Sequence Number (BE): 2 (0x0002)
Sequence Number (LE): 512 (0x0200)
[Request frame: 260]
[Response time: 69,727 ms]
Data (32 bytes)

ICMP dan Traceroute

1. c:\windows\system32\tracert www.inria.fr

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\fairuuz>c:\windows\system32\tracert www.inria.fr

Tracing route to inria.fr [128.93.162.83]
over a maximum of 30 hops:

  0  33 ms  42 ms  6 ms  192.168.0.1
  1  18 ms  6 ms  4 ms  192.168.1.1
  2  24 ms  13 ms  21 ms  10.114.0.1
  3  26 ms  8 ms  5 ms  180.250.252.69
  4  *  41 ms  *  180.240.190.77
  5  53 ms  58 ms  *  180.240.190.77
  6  187 ms  179 ms  180 ms  180.240.192.233
  7  198 ms  193 ms  193 ms  180.240.196.1
  8  273 ms  297 ms  290 ms  renater-par.franceix.net [37.49.236.19]
  9  192 ms  192 ms  196 ms  hu0-4-0-1-ren-nr-orsay-rtr-091.noc.renater.fr [193.51.180.43]
 10  229 ms  209 ms  198 ms  inria-rocquencourt-vl1631-tel-4-inria-rtr-021.noc.renater.fr [193.51.184.177]
 11  291 ms  283 ms  287 ms  unit240-reth1-vfw-ext-dcl.inria.fr [192.93.122.19]
 12  193 ms  197 ms  193 ms  prod-inriafr-cms.inria.fr [128.93.162.83]

Trace complete.

C:\Users\fairuuz>
```

2. ICMP after c:\windows\system32\tracert www.inria.fr in CMD

The image shows a Wireshark packet capture of ICMP Echo (ping) requests and responses. The packet list shows several 'Time to live exceeded' and 'Destination unreachable' messages. The packet details pane shows the structure of an ICMP Echo request.

No.	Source	Destination	Protocol	Length	Info
85	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=51/13056, ttl=1 (no response found!)
86	192.168.0.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
87	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=52/13312, ttl=1 (no response found!)
88	192.168.0.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
89	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=53/13568, ttl=1 (no response found!)
90	192.168.0.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
100	192.168.0.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)
106	192.168.0.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)
123	192.168.0.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)
138	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=54/13824, ttl=2 (no response found!)
140	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=55/14080, ttl=2 (no response found!)
142	192.168.1.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
143	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=56/14336, ttl=2 (no response found!)
144	192.168.1.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
148	192.168.1.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)
151	192.168.1.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)
157	192.168.1.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)

Frame 85: Packet, 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface Ethernet II, Src: LiteonTechno af:cf:7d (24:b2:b9:af:cf:7d), Dst: TPLink 5b:1b:b4 (bc:07:1d:5b:1b:b4)

Internet Protocol Version 4, Src: 192.168.0.102, Dst: 128.93.162.83

Internet Control Message Protocol

3. Detail Time-To-Live Exceeded

The image shows a Wireshark packet capture window titled "Wi-Fi". The main pane displays a list of captured packets, with the "icmp" filter applied. The packets are ICMP Echo (ping) requests and responses. The "Time-to-Live exceeded" error is highlighted in the packet list, indicating that the TTL has been exceeded during transit.

No.	Time	Source	Destination	Protocol	Length	Info
85	12.2340709..	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=51/13056, ttl=1 (no response found!)
86	12.2445549..	192.168.0.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
87	12.2452789..	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=52/13312, ttl=1 (no response found!)
88	12.2464346..	192.168.0.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
89	12.2468882..	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=53/13568, ttl=1 (no response found!)
90	12.2485626..	192.168.0.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
100	12.6682478..	192.168.0.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)
106	14.1887911..	192.168.0.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)
123	15.7099131..	192.168.0.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)
139	18.2256972..	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=54/13824, ttl=2 (no response found!)
139	18.2401958..	192.168.1.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
140	18.2477031..	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=55/14080, ttl=2 (no response found!)
142	18.2530041..	192.168.1.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
143	18.2533691..	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=56/14336, ttl=2 (no response found!)
144	18.2579009..	192.168.1.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
148	18.2756217..	192.168.1.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)
151	19.7698834..	192.168.1.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)
157	21.2844177..	192.168.1.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)

The packet details pane shows the selected packet (Frame 86) and its structure:

- Frame 86: Packet, 134 bytes on wire (1072 bits), 134 bytes captured (1072 bits) on interface
- Ethernet II, Src: TPLink_5b:1b:b4 (bc:07:1d:5b:1b:b4), Dst: LiteonTechno_af:cf:7d (24:b2:b
- Internet Protocol Version 4, Src: 192.168.0.1, Dst: 128.93.162.83
- Internet Control Message Protocol
 - Type: Time-to-live exceeded (11)
 - Code: 0 (Time to live exceeded in transit)
 - Checksum: 0xf4ff [correct]
 - [Checksum Status: Good]
 - Unused: 00000000
 - Internet Protocol Version 4, Src: 192.168.0.102, Dst: 128.93.162.83
 - Internet Control Message Protocol

The packet bytes pane shows the raw data of the packet, including the Ethernet II header, IP header, and ICMP header.

Internet Control Message Protocol: Protocol

Packets: 669 - Displayed: 88 (13.2%) - Dropped: 0 (0.0%)

Profile: Default